

Faculty Vitae: Vivek Sharma

1. Education

- B. Tech. Textile Technology, Indian Institute of Technology Delhi, New Delhi, India, 2001
M.S. Polymer Science, University of Akron, Akron, OH, 2003
M.S. Chemical Engineering, Minor: Nonlinear Sciences & Chaos, Georgia Institute of Technology, Atlanta, GA 2006
Ph.D. Polymer Science & Engineering, Minor: Physics, Georgia Institute of Technology Atlanta, GA, 2008
Post-Doc Mechanical Engineering & Hatsopoulos Microfluids Lab, Massachusetts Institute of Technology Cambridge, MA 2012

2. Academic Experience

Department of Chemical Engineering, University of Illinois Chicago, Professor, 2025-present, Full-Time

Department of Chemical Engineering, University of Illinois Chicago, Associate Professor, 2019-2025, Full-Time

Department of Chemical Engineering, University of Illinois Chicago, Assistant Professor, 2012-2019, Full-Time

Visiting Professor Appointments

SIMM: Soft Matter Sciences and Engineering Laboratory, ESPCI Paris – PSL, France. Visiting Professor (Chaire Paris Sciences) Nov 2025-Dec 2025

Pritzker School for Molecular Engineering (PME). University of Chicago, Hyde Park, Chicago, IL. Visiting Associate Professor, Jan 2023-June 2023

Chemical Engineering, Stanford University, Palo Alto, CA, Visiting Associate Professor, Sept 2022-Dec 2022.

Institute for Molecular Engineering, University of Chicago, Chicago IL, Visiting Assistant Professor, Sept 2017- Dec 2017, Part-Time

3. Non-Academic Experience

Not applicable

4. Certifications or professional registrations

Not applicable

5. Current membership in professional organizations

American Institute of Chemical Engineers, Society of Rheology, American Physical Society (APS), Adhesion Society, American Chemical Society

6. Honors and awards

John H. Dillon Medal from the American Physical Society, 2023.

3M Non-tenured Faculty Award, 2019.

Best Presentation Award for talk on “Computational analysis of pinch-off dynamics and printability of simple and complex fluids”, *Flow-3D Users Meeting*, Santa Fe, NM 2017 (co-author Jelena Dinic).

Distinguished Young Rheologist Award, TA Instruments, 2015.

UIC College of Engineering (COE) Faculty Award For Teaching. 2025

UIC College of Engineering (COE) Faculty Award For Teaching. 2023

UIC College of Engineering (COE) Faculty Award For Advising. 2022

UIC College of Engineering (COE) Faculty Award For Teaching. 2017

UIC HOPE (Honoring Our Professor’s Excellence) Award. 2017.

7. Service activities

Member of the graduate committee, chemical engineering (2014-2016 & 2017-2019, 2021-2022)

Seminar organizer for the department of chemical engineering, (2015-2019)

Website committee for the department of chemical engineering (2015-current)

Chair, Molecular Engineering Strategic Committee, for chemical engineering (2016-current)

Faculty search committee (2015-2016 & 2019-2020)

Outside UIC

Member-at-Large, APS-DPOLY, 2023-2026

Chair, SOR Centennial Committee, Society of Rheology (2022-2024)

Member, SOR Centennial Committee, Society of Rheology (2024-)

Member, Education Committee, APS DPOLY (2022-2023)

Member, ACS Colloids Summer Symposium Organizing Committee (2022-2028)

Publicity committee, APS DPOLY (2022-)

Chair, Membership Committee, GSOF (2016-2020); Helped in GSOF-DSOF transition!

Membership Committee, Society of Rheology (2017-2022)

Social Media Chair, Society of Rheology (2017-2022)

Programming Committee, Area 1J: Fluids, AIChE (2017-2025)

Reviewer: Soft Matter, Journal of Fluid Mechanics, Physical Review Letters, Advanced Materials, Journal of Rheology, Rheologica Acta, Physical Review E, International Journal of Multiphase Flow, Langmuir, Applied Rheology, International Journal of Polymer Science, Journal of Material Chemistry, Macromolecules, Lab Chip, Polymer Chemistry, RSC Advances, and Experiments in Fluids

Short Course Organizer or Instructor/ Workshop Organizer

Summer School on Soft Matter Systems: from Fundamentals to Sensory Perception, Instructor and co-

organizer of demos, Cargese, Corsica, France, July 2026

Workshop on Molecular Engineering of Soft Matter, co-organizer with Shikha Nangia, and Harm-Anton Klok. Telluride, May 2025.

Summer School on Soft Matter Systems: from Fundamentals to Food, Instructor and co-organizer of demos, Cargese, Corsica, France, July 2024

Workshop on Molecular Engineering of Soft Matter, co-organizer with Shikha Nangia, and Harm-Anton Klok. Telluride, June 2023.

Short Course on Macromolecular Engineering of Formulations, co-organized with Thomas O'Connor, under the aegis of DPOLY at the *APS March Meeting, 2021*.

Short course on Structure and Rheology of Emulsions and Foams co-organized and co-taught with Sibani Lisa Biswal, at the *Society of Rheology Meeting*, Houston, TX.

8. Selected Publications

M-A. Fardin, M. Hautefeuille, and V. Sharma, “Radical scaling: Beyond feet and fingers”, *Soft Matter*, 2025. <https://doi.org/10.48550/arXiv.2507.02631>

L. Hassan, K. Al Zahabi[#], N.N. Nikolova, M. W. Boehm, S. K. Baier, and V. Sharma, “Rheology of animal and plant milks featuring dripping-onto-substrate rheometry”, *Phys. Fluids* (2025).

R. B. Reboucas, N. N. Nikolova, V. Sharma, “Modeling drop deformations and rheology of dilute to dense emulsions”, *Current Opinions in Colloid and Interface Science*, invited review, (2025).

C. Xu, Y. Zhang, and V. Sharma*, “Spatiotemporal mapping of nanotopography and thickness transitions of ultrathin foam films”, *Soft Matter*, 20, 3719-3727 (2024)

N. Nikolova[#], C. D. V. Martínez Narváez, L. Hassan, R. A. Nicholson, M. W. Boehm, S. K. Baier, and V. Sharma, “Rheology and dispensing of real and vegan mayo: the chickpea or egg problem”, *Soft Matter*, **19**, 9413-9427 (2023). (Collaboration with Motif)

L. N. Jimenez, C. D. V. Martínez Narváez, and V. Sharma*, “Solvent Properties Influence Rheology and Pinching Dynamics of Polyelectrolyte Solutions: Thickening the Pot with Glycerol and Cellulose Gum”, *Macromolecules*, 55, 8117-8132 (2022).

J. Merchiers[†], C. D. V. Martínez Narváez[†], C. Slykas, N.K. Reddy and V. Sharma*, “Evaporation and rheology chart a processability map for centrifugal force spinning”, *Macromolecules*, 54 (23), 11061-11073 (2021). (w/ Hasselt)

C. Martinez Narvaez, X. Yu, C. Wang, R. Rock, H. Sun, and V. Sharma, “Rheology and pinching dynamics of associative polysaccharide solutions”, *Macromolecules*, 54, 6372-6288 (2021). (w/ PPG)

C. Ochoa, S. Gao, S. Srivastava, and V. Sharma*, “Foam film stratification studies probe intermicellar interactions”, C. Ochoa, S. Guo, S. Srivastava, and V. Sharma*, *Proceedings of the National Academy of Sciences of the USA*, 118 (25), e2024805118 (2021). (w/ UCLA)

J. Dinic and V. Sharma*, “Macromolecular Relaxation, Strain and Finite Extensibility Determine Visco-elastocapillary Thinning and Extensional Viscosity of Polymer Solutions”, *Proceedings of National Academy of Sciences*, 116(18), 8766-8774 (2019).

9. Professional Development

Chaired /organized sessions (>30) at ACS Colloids, APS-DFD, AIChE, SOR, APS March, ACS, and Adhesion Society Meetings.